🟢 Basic Level

1\. Take array input from user and print it

Problem: Write a program that takes an integer `n` as input, followed by `n` integers, stores them in an array, and then prints the elements of the array.

  * **Input Example:**
    ```
    5
    1 2 3 4 5
    ```
  * **Output Example:**
    ```
    1 2 3 4 5
    ```

2\. Print array in reverse order

Problem: Given an array `nums`, print its elements in reverse order.

  * **Input Example:**
    ```
    [1, 2, 3, 4, 5]
    ```
  * **Output Example:**
    ```
    5 4 3 2 1
    ```

3\. Find the sum of all array elements

Problem: Given an array of integers `nums`, find and print the sum of all its elements.

  * **Input Example:**
    ```
    [10, 20, 30, 40]
    ```
  * **Output Example:**
    ```
    100
    ```

4\. Find the maximum and minimum element

Problem: Given an array of integers `nums`, find and print the maximum and minimum elements in the array.

  * **Input Example:**
    ```
    [5, 8, 2, 10, 3]
    ```
  * **Output Example:**
    ```
    Maximum: 10
    Minimum: 2
    ```

5\. Count even and odd numbers in an array

Problem: Given an array of integers `nums`, count and print the total number of even and odd elements.

  * **Input Example:**
    ```
    [1, 2, 3, 4, 5, 6]
    ```
  * **Output Example:**
    ```
    Even count: 3
    Odd count: 3
    ```

6\. Print elements at even and odd indices

Problem: Given an array of integers `nums`, print all elements located at even indices and all elements located at odd indices.

  * **Input Example:**
    ```
    [10, 20, 30, 40, 50]
    ```
  * **Output Example:**
    ```
    Elements at even indices: 10 30 50
    Elements at odd indices: 20 40
    ```

```

#### 7\. Calculate average of array elements

**Problem:** Given an array of integers `nums`, calculate and print the average of its elements.

  * **Input Example:**
    ```
    [10, 20, 30]
    ```
  * **Output Example:**
    ```
    20.0
    ```

#### 8\. Search for an element in array (Linear Search)

**Problem:** Given an array of integers `nums` and a target integer `x`, find if `x` is present in the array. Return the index of `x` if found, otherwise return -1.

  * **Input Example:**
    ```
    nums = [10, 20, 30, 40], x = 30
    ```
  * **Output Example:**
    ```
    2
    ```

#### 9\. Count the frequency of each element

**Problem:** Given an array of integers `nums`, count and print the frequency of each distinct element in the array.

  * **Input Example:**
    ```
    [1, 2, 2, 3, 1, 4, 2]
    ```
  * **Output Example:**
    ```
    1: 2
    2: 3
    3: 1
    4: 1

```

#### 10\. Copy array elements to another array

**Problem:** Given an array `nums`, create a new array `new_nums` of the same size and copy all elements from `nums` to `new_nums`. Print the new array.

  * **Input Example:**
    ```
    [1, 5, 9, 13]
    ```
  * **Output Example:**
    ```
    [1, 5, 9, 13]
    ```

#### 11\. Find the largest element in the array

**Problem:** Given an array of integers `nums`, find and print the largest element.

  * **Input Example:**
    ```
    [2, 7, 1, 9, 5]
    ```
  * **Output Example:**
    ```
    9
    ```

#### 12\. Check if the array is sorted (ascending order)

**Problem:** Given an array of integers `nums`, return `true` if the array is sorted in ascending order, otherwise return `false`.

  * **Input Example:**
    ```
    [1, 2, 3, 5]
    ```
  * **Output Example:**
    ```
    true
    ```

#### 13\. Find the second largest and second smallest element
    ```

Problem: Given an array of integers `nums` with at least two elements, find and print the second largest and second smallest distinct elements.

  * **Input Example:**
    ```
    [5, 2, 8, 1, 9]
    ```
  * **Output Example:**
    ```
    Second largest: 8
    Second smallest: 2
    ```

14\. Find the kth largest and kth smallest element

Problem: Given an array of integers `nums` and an integer `k`, find and print the kth largest and kth smallest elements in the array.

  * **Input Example:**
    ```
    nums = [3, 2, 1, 5, 6, 4], k = 2
    ```
  * **Output Example:**
    ```
    2nd largest: 5
    2nd smallest: 2
    ```

🟡 Intermediate Level

15\. Left rotate array by 1 position

Problem: Given an array of integers `nums`, rotate the array to the left by one position. The first element should move to the end of the array.

  * **Input Example:**
    ```
    [1, 2, 3, 4, 5]
    ```
  * **Output Example:**
    ```
    [2, 3, 4, 5, 1]
    ```

16\. Right rotate array by 1 position

Problem: Given an array of integers `nums`, rotate the array to the right by one position. The last element should move to the beginning of the array.

  * **Input Example:**
    ```
    [1, 2, 3, 4, 5]
    ```
  * **Output Example:**
    ```
    [5, 1, 2, 3, 4]
    ```

17\. Reverse an array in place

Problem: Given an array of integers `nums`, reverse the array without using any extra space.

  * **Input Example:**
    ```
    [1, 2, 3, 4]
    ```
  * **Output Example:**
    ```
    [4, 3, 2, 1]
    ```

18\. Rotate array by k positions (left/right)

Problem: Given an array of integers `nums` and an integer `k`, rotate the array to the left by `k` positions.

  * **Input Example:**
    ```
    nums = [1, 2, 3, 4, 5], k = 2
    ```
  * **Output Example:**
    ```
    [3, 4, 5, 1, 2]
    ```

19\. Remove duplicates from sorted array

Problem: Given a sorted array `nums`, remove the duplicates in-place such that each unique element appears only once. Return the new length of the array.

  * **Input Example:**
    ```
    [1, 1, 2, 2, 3, 4]
    ```
  * **Output Example:**
    ```
    4 (The array becomes [1, 2, 3, 4] after modification)
    ```

20\. Find the missing number in 1 to n

Problem: Given an array `nums` containing `n` distinct numbers taken from `0, 1, 2, ..., n`, find the single number that is missing.

  * **Input Example:**
    ```
    [3, 0, 1]
    ```
  * **Output Example:**
    ```
    2
    ```

21\. Two Sum problem (find pair with given sum)

Problem: Given an array of integers `nums` and an integer `target`, return indices of the two numbers such that they add up to `target`.

  * **Input Example:**

```
nums = [2, 7, 11, 15], target = 9
```
* **Output Example:**
```
[0, 1]
```

22\. Find all pairs with given sum

Problem: Given an array of integers `nums` and an integer `target`, find and print all unique pairs of numbers that sum up to `target`.

* **Input Example:**
```
nums = [1, 5, 7, -1, 5], target = 6
```
* **Output Example:**
```
(1, 5)
(7, -1)
```

23\. Majority element (\> n/2 times)

Problem: Given an array `nums` of size `n`, return the majority element, which is the element that appears more than `⌊n / 2⌋` times.

* **Input Example:**
```
[3, 2, 3]
```
* **Output Example:**
```
3
```

24\. Find element appearing more than n/3 times

Problem: Given an integer array of size `n`, find all elements that appear more than `⌊n / 3⌋` times.

* **Input Example:**
```
```

[3, 2, 3, 1, 2, 2]
```
* **Output Example:**
```
[2]
```

#### 25\. Find equilibrium index (sum left = sum right)

**Problem:** Given a 0-indexed integer array `nums`, find the leftmost index `i` such that the sum of elements to the left of `i` is equal to the sum of elements to the right of `i`.

* **Input Example:**
```
[2, 3, -1, 8, 4]
```
* **Output Example:**
```
3
```

#### 26\. Rearrange array alternatively +ve and -ve

**Problem:** Given an array of positive and negative numbers, rearrange them so that positive and negative numbers alternate, while maintaining their relative order.

* **Input Example:**
```
[1, -2, 3, -4, 5, -6]
```
* **Output Example:**
```
[1, -2, 3, -4, 5, -6]
```

#### 27\. Leaders in an array (greater than all to right)

**Problem:** An element is a leader if it is greater than all the elements to its right. Given an array `nums`, find and print all its leaders.
```

* **Input Example:**
```
[16, 17, 4, 3, 5, 2]
```
* **Output Example:**
```
17, 5, 2
```

28\. Maximum difference arr[j] – arr[i] (j \> i)

Problem: Given an array `nums`, find the maximum value of `nums[j] - nums[i]` such that `j > i`.

* **Input Example:**
```
[7, 1, 5, 4, 3, 6]
```
* **Output Example:**
```
5 (from 6 - 1)
```

29\. Stock buy and sell to maximize profit

Problem: Given an array `prices` where `prices[i]` is the price of a given stock on the `i`th day, find the maximum profit you can achieve. You are allowed to complete at most one transaction.

* **Input Example:**
```
[7, 1, 5, 3, 6, 4]
```
* **Output Example:**
```
5 (buy on day 2 at price 1, sell on day 5 at price 6)
```

30\. Maximum subarray sum (Kadane’s Algorithm)

Problem: Given an integer array `nums`, find the contiguous subarray (containing at least one number) which has the largest sum and return its sum.

  * **Input Example:**
    ```
    [-2, 1, -3, 4, -1, 2, 1, -5, 4]
    ```
  * **Output Example:**
    ```
    6 (from subarray [4, -1, 2, 1])
    ```

31\. Triplets with sum = 0

Problem: Given an integer array `nums`, return all unique triplets `[a, b, c]` such that `a + b + c = 0`.

  * **Input Example:**
    ```
    [-1, 0, 1, 2, -1, -4]
    ```
  * **Output Example:**
    ```
    [[-1, -1, 2], [-1, 0, 1]]
    ```

32\. Intersection of two arrays

Problem: Given two integer arrays `nums1` and `nums2`, return an array of their intersection. Each element in the result should appear as many times as it shows in both arrays.

  * **Input Example:**
    ```
    nums1 = [1, 2, 2, 1], nums2 = [2, 2]
    ```
  * **Output Example:**
    ```
    [2, 2]
    ```

33\. Union of two arrays

Problem: Given two integer arrays `nums1` and `nums2`, return an array of their union. The union should contain each distinct element that appears in either or both arrays.

  * **Input Example:**
    ```
    nums1 = [1, 2, 3], nums2 = [2, 4, 5]
    ```
  * **Output Example:**
    ```
    [1, 2, 3, 4, 5]
    ```

34\. Check if array is palindrome

Problem: Given an array of integers `nums`, return `true` if the array is a palindrome, otherwise return `false`.

  * **Input Example:**
    ```
    [1, 2, 3, 2, 1]
    ```
  * **Output Example:**
    ```
    true
    ```

35\. Count positive, negative and zero elements

Problem: Given an array of integers `nums`, count and print the number of positive, negative, and zero elements.

  * **Input Example:**
    ```
    [1, -2, 0, 4, -5, 0]
    ```
  * **Output Example:**
    ```
    Positive: 2
    Negative: 2
    Zeros: 2
    ```

36\. Find sum of elements at even and odd indices separately

Problem: Given an array of integers `nums`, calculate and print the sum of all elements at even indices and the sum of all elements at odd indices separately.

  * **Input Example:**
    ```
    [1, 2, 3, 4, 5]
    ```
  * **Output Example:**
    ```
    Sum of even indices: 9
    Sum of odd indices: 6
    ```

🔵 Advanced Level (Binary Search / Optimized)

37\. Search element in sorted rotated array

Problem: Given a sorted array that has been rotated at some pivot, and a target value, search for the target in the array.

  * **Input Example:**
    ```
    nums = [4, 5, 6, 7, 0, 1, 2], target = 0
    ```

```
* **Output Example:**
```
4
```

#### 38\. Find peak element

**Problem:** A peak element is an element that is strictly greater than its neighbors. Given an integer array `nums`, find a peak element and return its index.

* **Input Example:**
```
[1, 2, 1, 3, 5, 6, 4]
```
* **Output Example:**
```
1 (since nums[1] is 2) or 5 (since nums[5] is 6)
```

#### 39\. Find first and last occurrence of element

**Problem:** Given a sorted array `nums` with duplicates, and a target integer `target`, find the first and last occurrence of `target`.

* **Input Example:**
```
nums = [5, 7, 7, 8, 8, 10], target = 8
```
* **Output Example:**
```
[3, 4]
```

#### 40\. Count occurrences of an element

**Problem:** Given a sorted array `nums` and a target integer `target`, count the number of times `target` appears in the array.

* **Input Example:**
```
nums = [1, 2, 2, 2, 3, 4], target = 2

```
  * **Output Example:**
    ```
    3
    ```

#### 41\. Find floor and ceil of an element

**Problem:** Given a sorted array `nums` and a target value `x`, find the **floor** of `x` (largest element ≤ `x`) and the **ceil** of `x` (smallest element ≥ `x`).

  * **Input Example:**
    ```
    nums = [1, 2, 8, 10, 11, 12, 19], x = 5
    ```
  * **Output Example:**
    ```
    Floor of 5 is 2.
    Ceil of 5 is 8.
    ```

#### 42\. Find position to insert element in sorted array

**Problem:** Given a sorted array `nums` and a target value `target`, return the index where the `target` would be if it were inserted in order.

  * **Input Example:**
    ```
    nums = [1, 3, 5, 6], target = 5
    ```
  * **Output Example:**
    ```
    2
    ```

#### 43\. Find index of element in infinite sorted array

**Problem:** Given an array with an infinite number of elements, sorted in ascending order, and a target value, find the index of the target.

  * **Input Example:**
    ```

   nums = [3, 5, 7, 9, 10, 90, 100, 130, 140, 160, 170, ...], target = 10
   ```
 * **Output Example:**
   ```
   4
   ```

#### 44\. Find minimum in rotated sorted array

**Problem:** Given a sorted array that has been rotated, find the minimum element. The array contains unique elements.

 * **Input Example:**
   ```
   [3, 4, 5, 1, 2]
   ```
 * **Output Example:**
   ```
   1
   ```

#### 45\. Search in 2D sorted matrix

**Problem:** Given a 2D matrix where each row and each column are sorted, search for a target value. Return `true` if the target is in the matrix, otherwise return `false`.

 * **Input Example:**
   ```
   matrix = [[1, 4, 7], [2, 5, 8], [3, 6, 9]], target = 5
   ```
 * **Output Example:**
   ```
   true
   ```
   ```